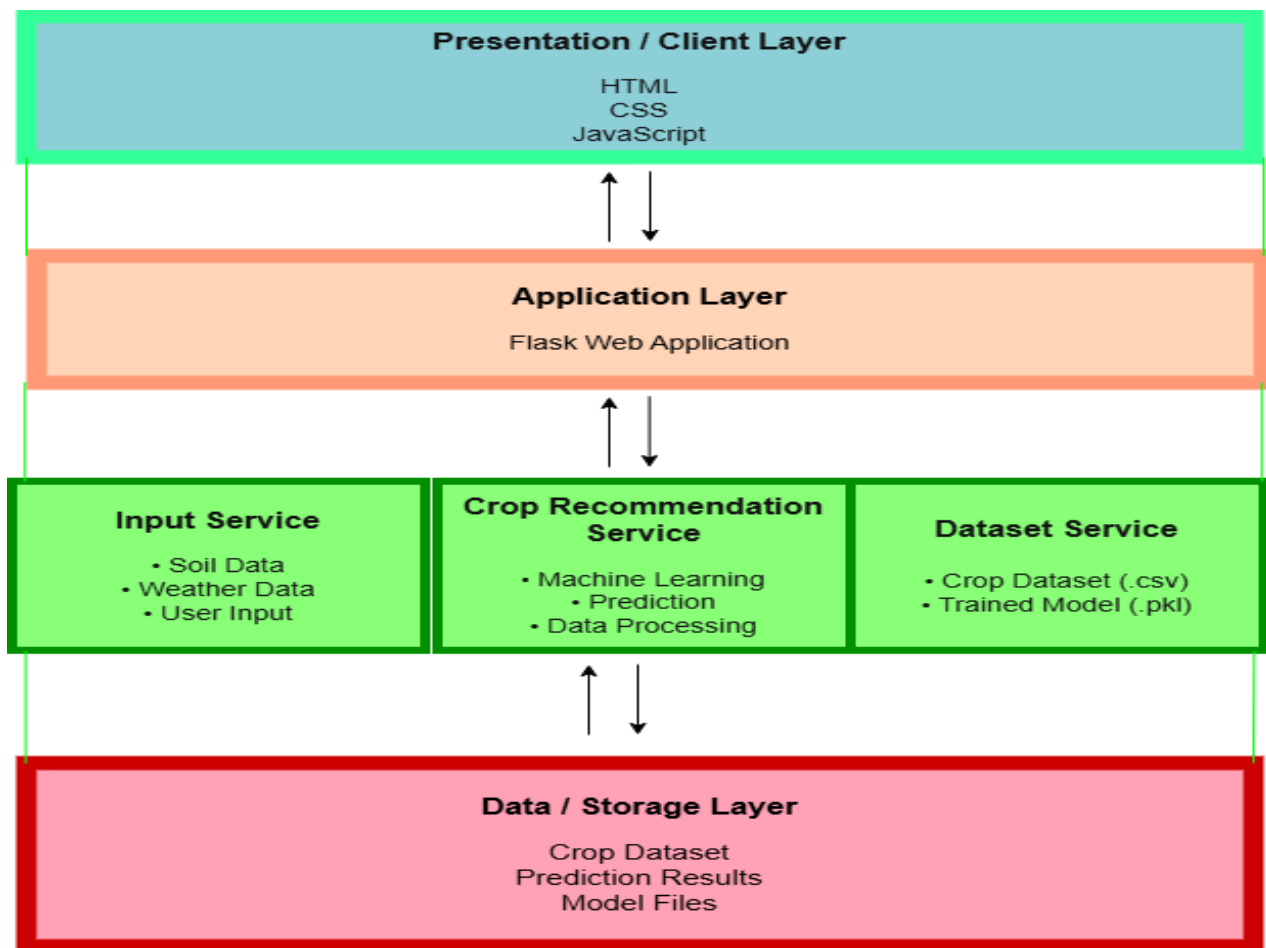


Solution Architecture

Date	27 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The OptiCrop system follows a layered architecture that separates the presentation, application, machine learning, and data storage components. Farmers interact with the web interface to provide soil and weather details. The Flask backend processes the input, communicates with the machine learning model, and retrieves the predicted crop recommendation from the trained model and dataset. This architecture ensures modularity, scalability, and efficient processing.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides the user interface to enter soil and weather details and display crop recommendations.	HTML, CSS, JavaScript
Application Layer	Handles requests, processes user input, and connects the frontend with the ML model.	Flask
Input Service	Collects soil, weather, and user input for crop prediction.	Python

Crop Recommendation Service	Processes input data and predicts the best crop using the trained machine learning model.	Python, Scikit-learn
Dataset Service	Stores the crop dataset and trained machine learning model used for prediction.	CSV Dataset, Pickle (.pkl)
Data / Storage Layer	Stores datasets, trained models, and prediction results.	CSV Dataset, Pickle (.pkl)